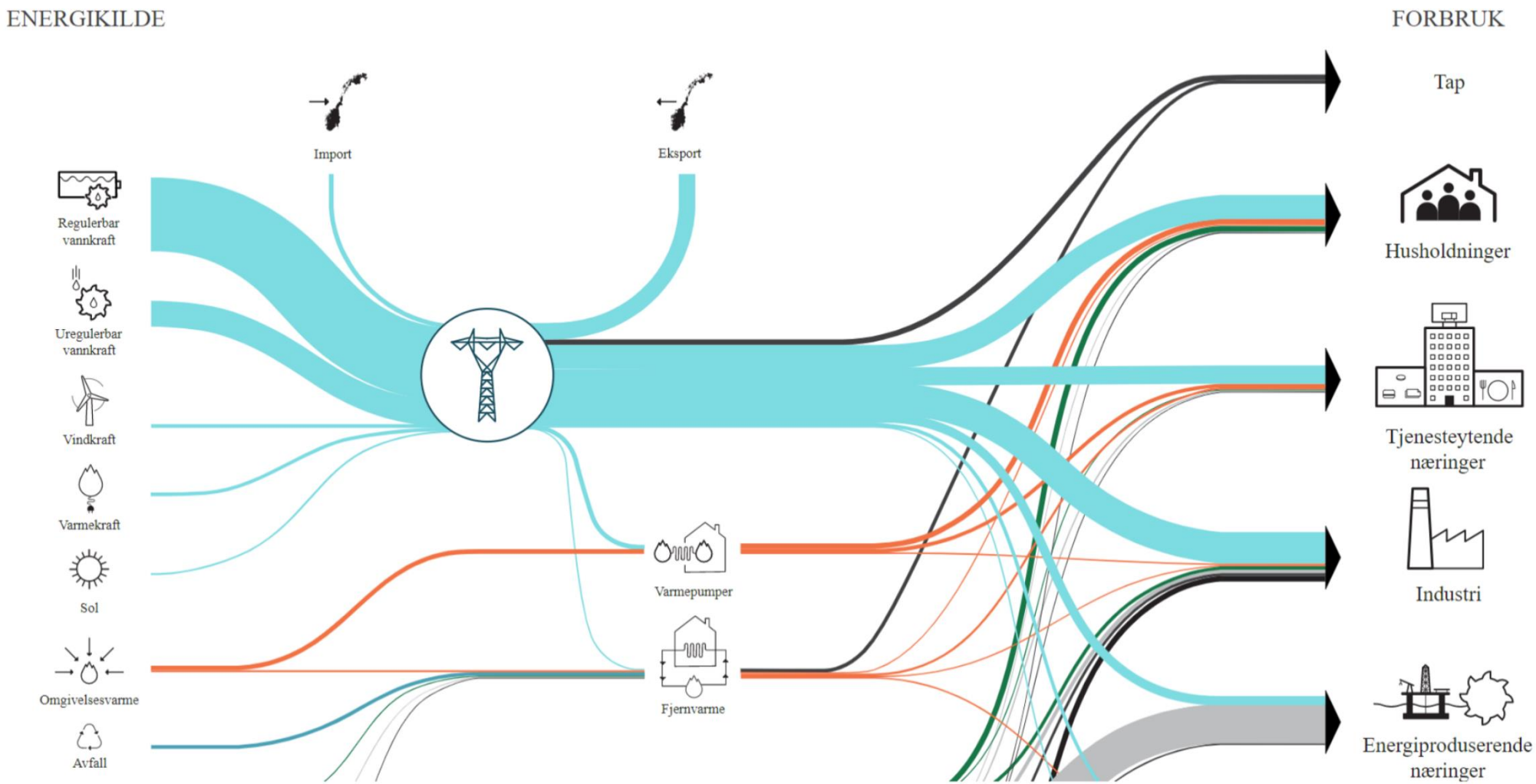
Day-ahead energy consumption forecasting

Aneo

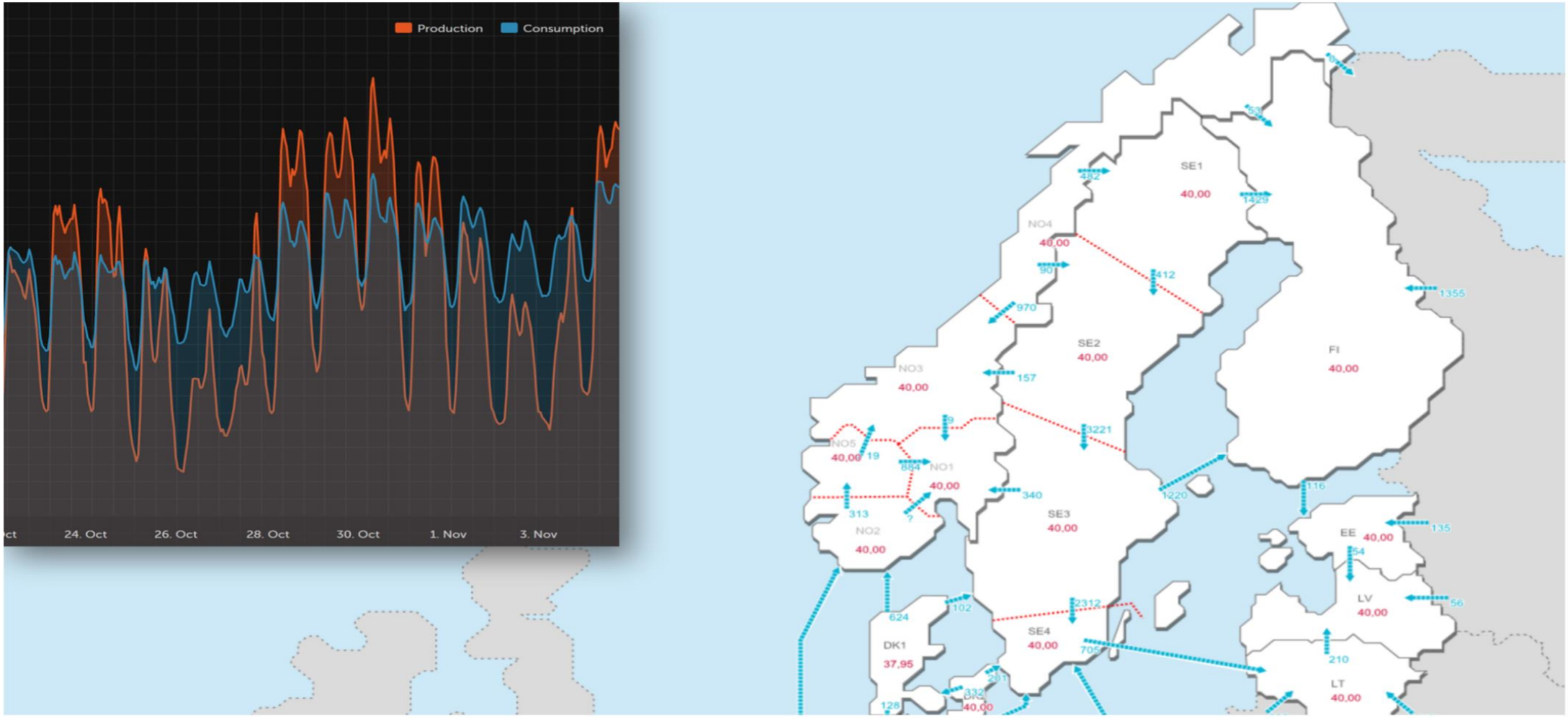
Nisha Dalal

Energy system

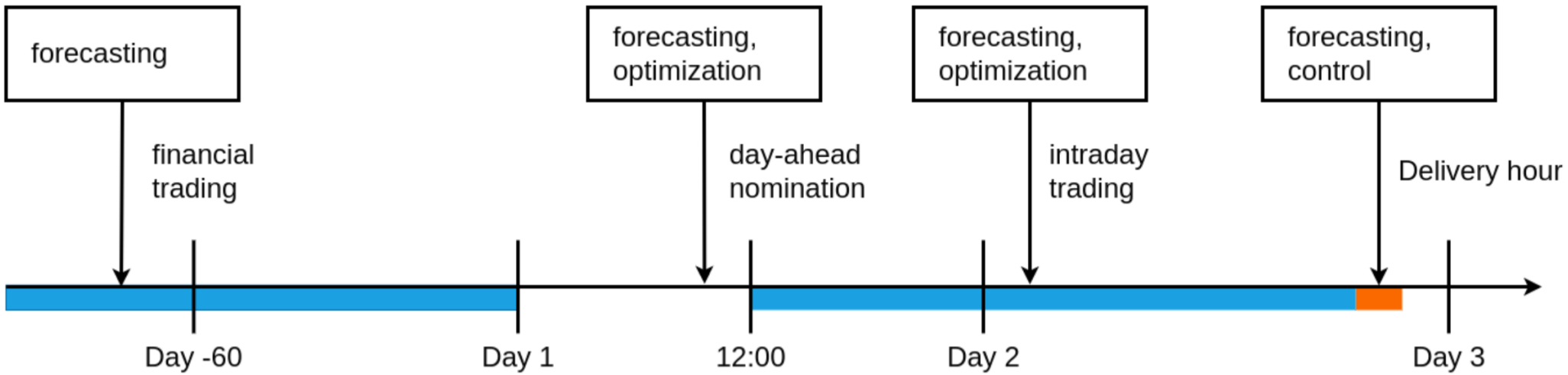


Balancing

Production
Consumption
24. Oct
26. Oct
28. Oct
30. Oct
1. Nov
3. Nov


SE1
40,00
NO4
40,00
SE2
40,00
NO3
40,00
FI
40,00
NO5
40,00
NO1
40,00
SE3
40,00
NO2
40,00
EE
40,00
LV
40,00
SE4
40,00
DK1
37,95
LT
40,00
482
1429
90
412
970
1355
157
3221
19
854
340
1220
313
116
135
54
624
102
2312
56
705
210
201
832
40,00
128

Energy management timeline

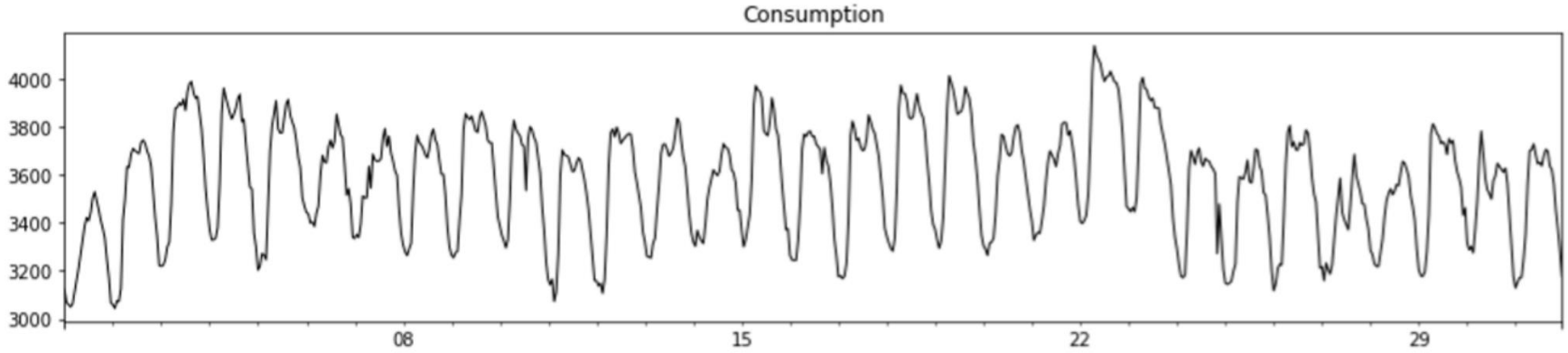


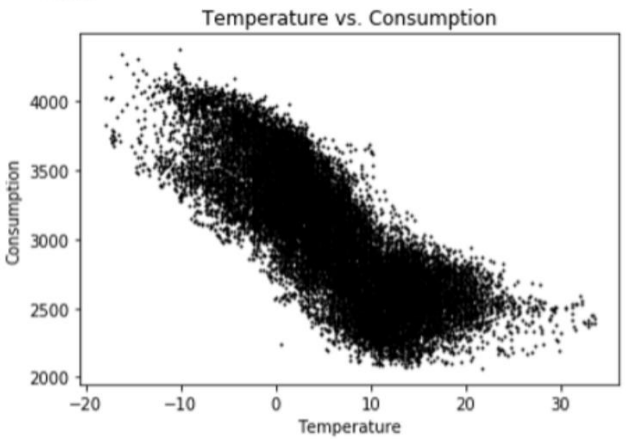
Project: Day-ahead consumption forecasting

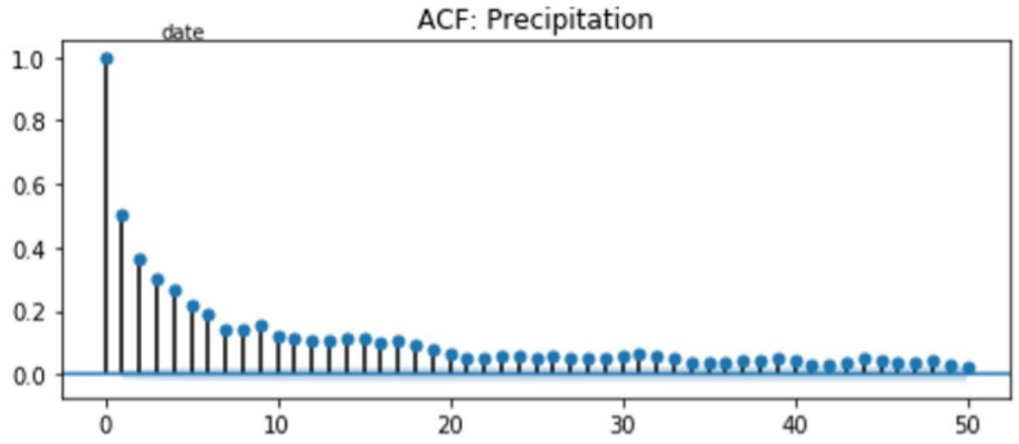
- Day-ahead forecast is made every day at 11:00 for all hours of the next day (24 values).
- Consumption for Aneo clients in 5 cities, named by city name.
- Develop and document a forecasting pipeline including data preparation, model training and evaluation.
- Explain the choice of methods used in the pipeline.

> Don't trust ChatGPT, it recommends methods that perform poorly on this task.

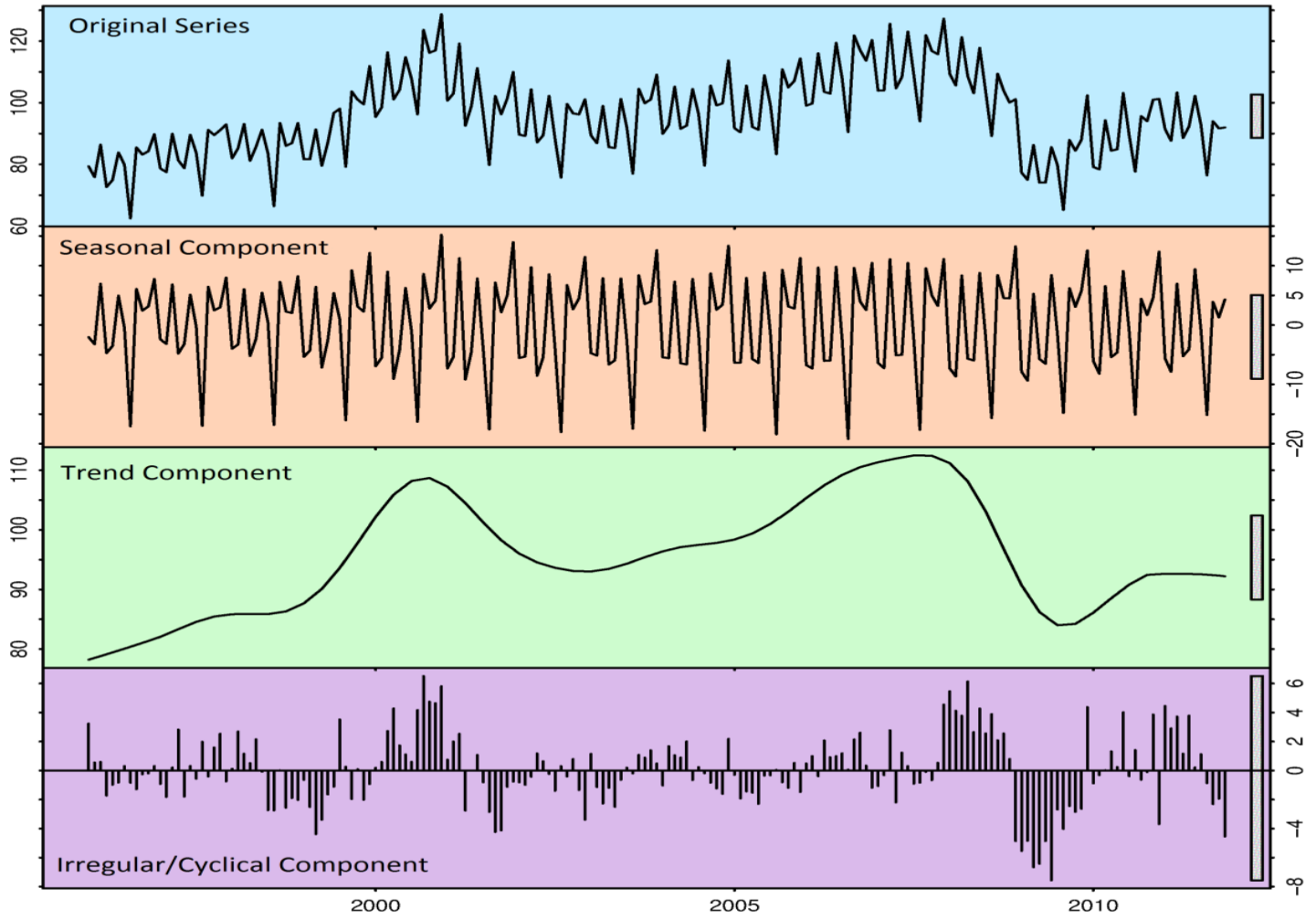
Data


Consumption
4000
3800
3600
3400
3200
3000
08
15
22
29
Jan
2018

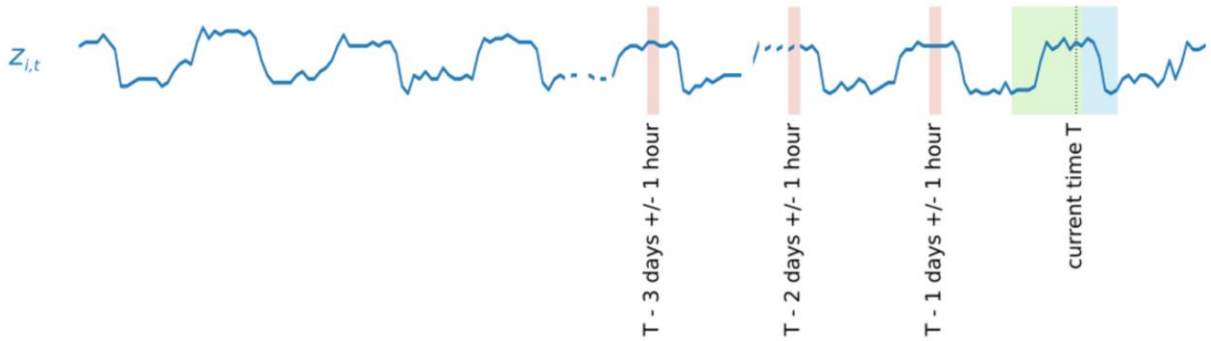

Temperature vs. Consumption
Consumption
4000
3500
3000
2500
2000
−20
−10
0
10
20
30
Temperature


ACF: Precipitation
date
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
50

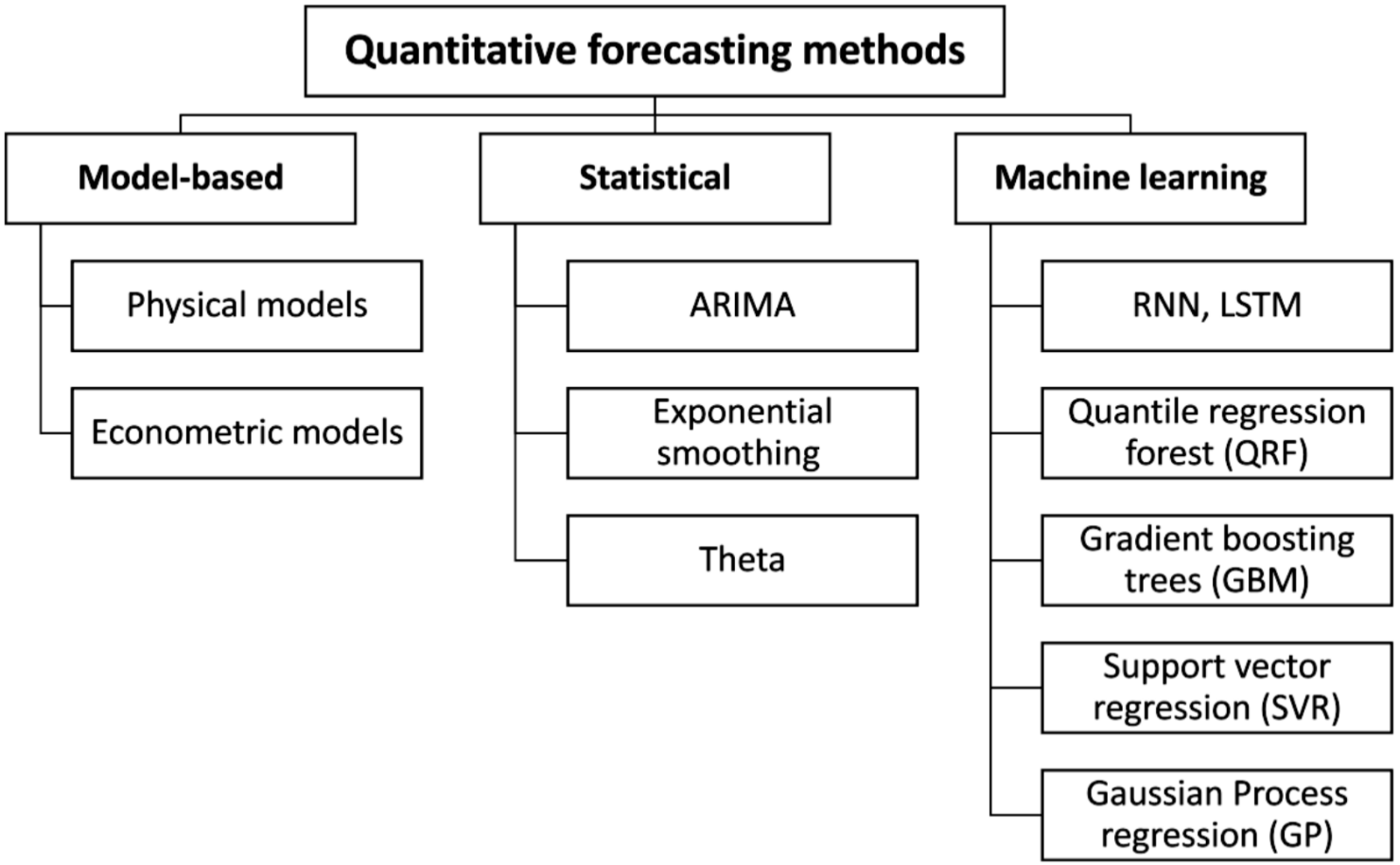
Time series components



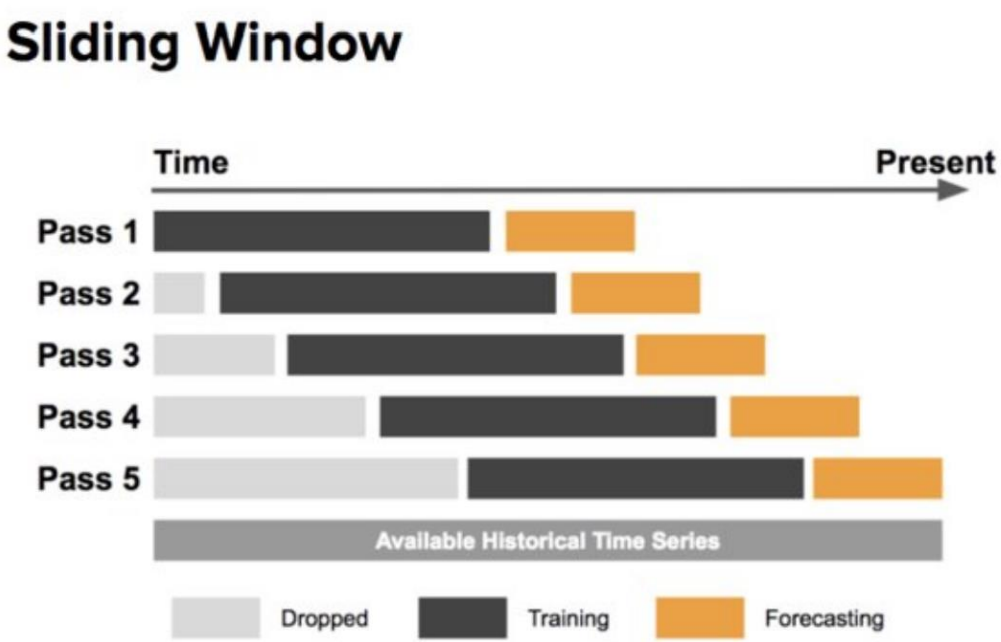
Lag features

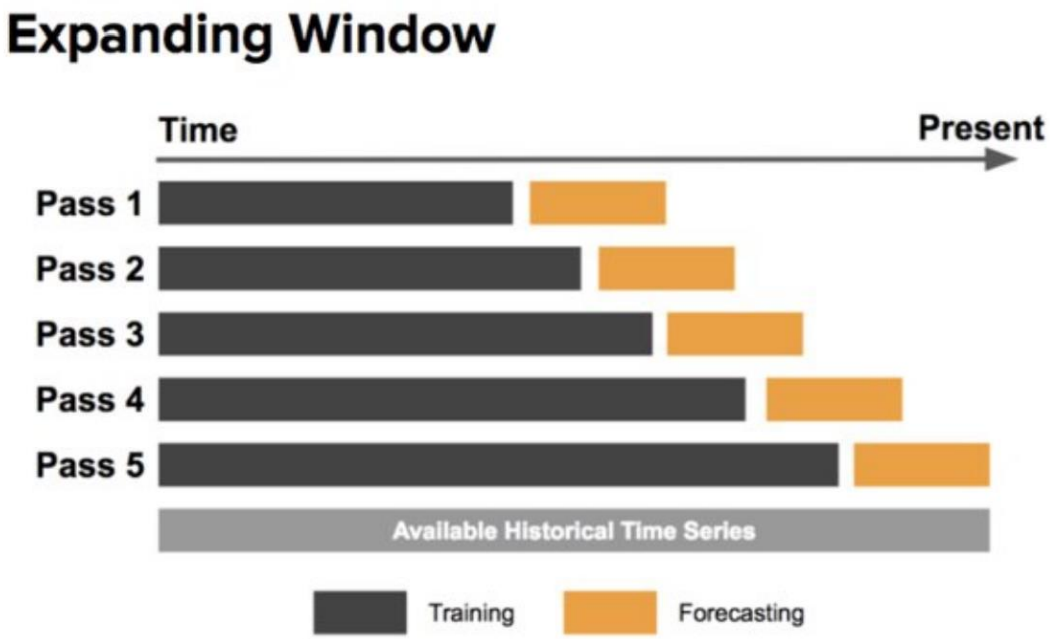


Forecasting models



Validation


Sliding Window
Time
Present
Pass 1
Pass 2
Pass 3
Pass 4
Pass 5
Available Historical Time Series
Dropped
Training
Forecasting


Expanding Window
Time
Present
Pass 1
Pass 2
Pass 3
Pass 4
Pass 5
Available Historical Time Series
Training
Forecasting

Forecasting pipeline

1. Domain understanding
2. Data acquisition
3. Data exploration and visualization
4. Data preprocessing: cleaning, transformation, normalization
5. Feature engineering: construction, selection
6. Model engineering: construction, selection, hyperparameter tuning
7. Evaluation: cross-validation, comparing with baselines

Project: Day-ahead consumption forecasting

- Day-ahead forecast is made every day at 11:00 for all hours of the next day (24 values).
- Consumption for Aneo clients in 5 cities, named by city name.
- Develop and document a forecasting pipeline including data preparation, model training and evaluation.
- Explain the choice of methods used in the pipeline.

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ANEO